La Scuola d'Italia / Dr. Huson / IB Math
18 April 2026

First & Last name:

5.14 Classwork: Logarithms and Exponentials (no calculator)

1. [Maximum mark: 5]

   Consider the curve with equation $y = (2x-1)e^{kx}$, where $x \in \mathbb{R}$ and $k \in \mathbb{R}$.

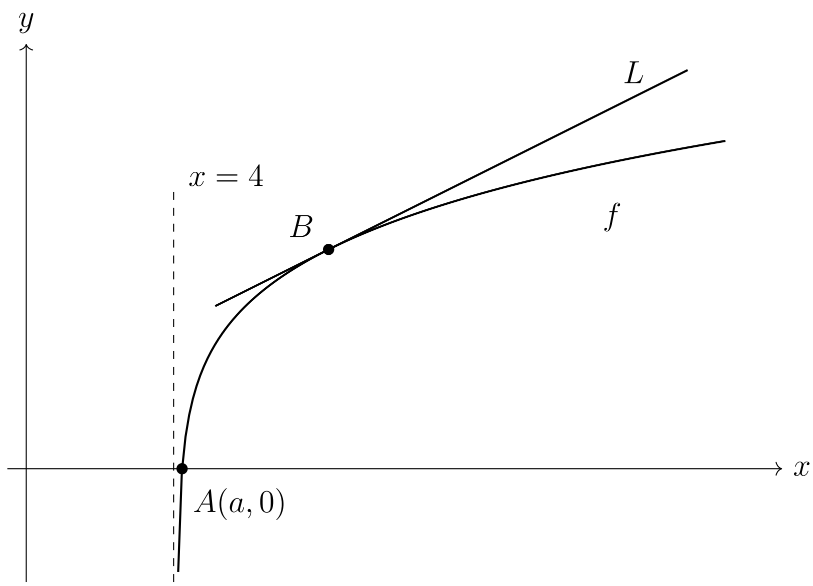
   The tangent to the curve at the point where $x = 1$ is parallel to the line $y = 5xe^k$.

   Find the value of k.

2. [Maximum mark: 9]

Consider the function f defined by $f(x) = \ln(x^2 - 16)$ for $x > 4$.

The following diagram shows part of the graph of f which crosses the x-axis at point A, with coordinates $(a, 0)$. The line L is the tangent to the graph of f at the point B.



(a) Find the exact value of a. [3]

(b) Given that the gradient of L is $\frac{1}{3}$, find the x-coordinate of B. [6]

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3. [Maximum mark: 14]

Let $g(x) = p^x + q$, for x, p, $q \in \mathbb{R}$, $p > 1$. The point $A(0, a)$ lies on the graph of g.

Let $f(x) = g^{-1}(x)$. The point B lies on the graph of f and is the reflection of point A in the line $y = x$.

(a) Write down the coordinates of B. [2]

The line L_1 is tangent to the graph of f at B.

(b) Given that $f'(a) = \dfrac{1}{\ln p}$, find the equation of L_1 in terms of x, p and q. [5]

The line L_2 is tangent to the graph of g at A and has equation $y = (\ln p)\, x + q + 1$.

The line L_2 passes through the point $(-2, -2)$.

The gradient of the normal to g at A is $\dfrac{1}{\ln \frac{1}{3}}$.

(c) Find the equation of L_1 in terms of x. [7]

Continue on lined paper.